

MPLADS AI Monitor

Risk Scoring Model: Complete Technical Explanation

Generated from the implementation in the mplads-ai-monitor project

Executive Summary

This system is not a supervised fraud-probability model. The official data contains no confirmed fraud labels, so the system cannot learn fraud probability. It is an explainable statistical risk engine that measures how unusual a work is compared with similar official records and how strongly the available data supports that comparison.

The final work score is a 0-100 composite made from six explainable components, with an unsupervised anomaly ensemble contributing only corroborating evidence.

End-to-End Pipeline

Official MPLADS data is cleaned and normalized, transformed into work, MP, payment, district, and vendor features, analyzed for duplicate descriptions, passed through anomaly detectors, converted into six risk components, and finally combined into a composite score with a risk band and plain-language explanation.

```
official data -> cleaning -> feature engineering -> duplicate detection -> anomaly ensemble -> risk components -> composite score -> band and explanation
```

1. Peer Comparison and Robust Statistics

Works are compared with peer groups such as the same state and category, the same district, the same category nationally, and the national fallback distribution. Each peer group stores its median amount, median absolute deviation (MAD), 90th percentile, and record count.

```
robust z = 0.6745 * (amount - peer median) / MAD
```

The median and MAD are used instead of the mean and standard deviation because monetary data is heavy-tailed and can contain extreme values. A peer group with fewer than 20 records is considered thin, falls back to national statistics, and causes the cost score to be reduced by 40%.

The analysis layer also calculates IQR outliers, extreme IQR outliers, percentiles, skewness, kurtosis, Gini concentration, and HHI. IQR statistics are primarily exploratory and help understand the observed data distribution; the work cost score mainly uses robust z-scores and peer ratios.

2. Feature Engineering

- Amount features: $\log(\text{amount} + 1)$, peer median ratio, robust z-scores, round-number indicators.
- Timing features: work age and age relative to the median age of completed works.
- Text features: description length, token count, repeated-description counts.
- Financial features: missing-data ratio, MP utilization, pending payments, expenditure gaps.
- Concentration features: district vendor HHI, MP top-vendor share, vendor HHI, payment repetition.

3. NLP Duplicate Detection

The duplicate detector uses normalized work descriptions, character-level TF-IDF, cosine similarity, and nearest-neighbor search. It uses character n-grams from 3 to 5 characters with a char_wb analyzer, which helps tolerate spelling variation, abbreviations, and word-order differences.

```
cosine similarity = (A dot B) / (||A|| * ||B||)
```

Setting	Value
Near-duplicate threshold	0.85 similarity
Exact duplicate threshold	0.995 similarity
Comparison blocking	State, then district/chunks for large states
Maximum matches	10 per work

For every work, the system records the highest similarity, the number of similar partners, and whether a similar work belongs to the same MP. Similar descriptions are signals for review, not proof of wrongdoing.

4. Anomaly Detection Ensemble

Four independent detector families are run for batch records:

Detector	Purpose
Isolation Forest	Finds globally rare multivariate records using random isolation trees.
Local Outlier Factor	Finds records whose local density differs from nearby records.
DBSCAN	Finds records that belong to no sufficiently dense cluster; label -1 is noise.
Statistical rules	Flags transparent conditions such as high peer cost, round amounts, repeated descriptions, repeated payments,

Isolation Forest uses 300 trees, 3% contamination, and random seed 42. Its raw anomaly score is normalized to 0-100 using the fitted population's 1st and 99.9th percentiles. This value is not a fraud probability.

The statistical rule score is:

```
statistical score = min(100, 100 * triggered rules / available rules * 2)
```

The batch ensemble is:

```
ensemble = 0.35*IsolationForest + 0.25*LOF + 0.10*DBSCAN_noise + 0.30*statistical_rules
```

5. Six Explainable Risk Components

Component	Calculation
Cost risk	55% peer robust z-score + 30% amount/peer-median ratio + 15% district robust z-score. Thin peer groups are discarded.
Duplicate risk	55% top text similarity + 25% duplicate-partner count + 20% same-MP repetition. Same-MP repetition receives full weight.
Delay risk	For recommended works only: open-age ratio ramped from 1.0 to 4.0. Completed works receive zero delay risk.
Vendor risk	50% district vendor HHI + 30% MP top-vendor share + 20% MP repeated-payment share.
Utilisation risk	50% unspent percentage + 30% pending-payment share + 20% published-vs-derived expenditure gap.
Data-quality risk	40% missing-data ratio + 30% missing/short description + 30% repeated work ID.

Each input is converted to 0-100 using a piecewise-linear ramp:

```
ramp(x, lo, hi) = clip(100 * (x - lo) / (hi - lo), 0, 100)
```

6. Final Composite Risk Formula

Component	Weight
Cost risk	24%

Component	Weight
Duplicate risk	22%
Vendor risk	20%
Delay risk	14%
Utilisation risk	12%
Data-quality risk	8%

The weighted component score is:

$$W = 0.24C + 0.22D + 0.20V + 0.14T + 0.12U + 0.08Q$$

The strongest component is $M = \max(C, D, V, T, U, Q)$. The explained score is blended with that strongest component so one severe, well-supported issue cannot be hidden by several quiet components:

$$X = 0.60W + 0.40M$$

Finally, the anomaly ensemble is added as corroborating evidence:

$$\text{Composite risk} = 0.85X + 0.15 \cdot \text{ensemble}$$

The composite is clipped to the 0-100 range and rounded to two decimal places.

7. Risk Bands and Alerts

Score	Band
0-24	LOW
25-49	MODERATE
50-74	HIGH
75-100	CRITICAL

An alert is generated at a composite score of 60 or above. The alert means the record should be reviewed; it is not a finding of fraud.

8. Live API Scoring

The live endpoint is POST /api/score. State and amount are required. Description, category, MP, house, district, agency, date, and work stage improve the assessment.

A versioned scoring pack freezes the peer medians, MAD values, district concentration, MP context, typical completed-work age, and fitted Isolation Forest from the latest pipeline run. This allows a new record to be scored against the same reference population as the batch register.

LOF and DBSCAN cannot score a single record independently because they describe a population. For live scoring, the ensemble falls back to the Isolation Forest score where LOF is unavailable, and the response explicitly reports this limitation.

9. Separate Completion-Propensity Model

The project also contains a separate HistGradientBoostingClassifier. It predicts whether an open work resembles previously completed works. It uses numerical and categorical work features, ordinal encoding, and a GroupShuffleSplit by MP so the same MP does not appear in both training and test sets.

This produces completion_propensity. It is not included in composite_risk and is not a fraud score. The project deliberately does not model fraud probability, completion time, cost overrun, or payment failure because the official data does not provide reliable labels for those outcomes.

10. Important Integration Issue

The Node controller in backend/controller/ml.js currently returns {status: success} before forwarding the request to FastAPI. Therefore POST /predict currently reports success without calculating a risk score. The actual scoring endpoint is the Python API at POST http://localhost:8000/api/score.

Implementation References

- risk_engine/engine.py - component calculations, composite score, bands, explanations
- anomaly_detection/detectors.py - Isolation Forest, LOF, DBSCAN, statistical rules
- ml/feature_engineering/features.py - work, MP, and vendor features
- nlp/duplicate_detection.py - TF-IDF and duplicate matching
- risk_engine/scoring_pack.py - frozen reference data for live scoring
- backend/services/scoring.py - live-record scoring
- prediction/models.py - completion-propensity model
- config/config.yaml - thresholds, weights, bands, and alert settings

Disclaimer: the system reports statistical deviation from comparable official records. Legitimate works can score highly. A risk score is a review indicator, not evidence of fraud or wrongdoing.